

Data Flow Diagram

Date	15 March 2026
Team ID	xxxxxx
Project Name	xxxxxx
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

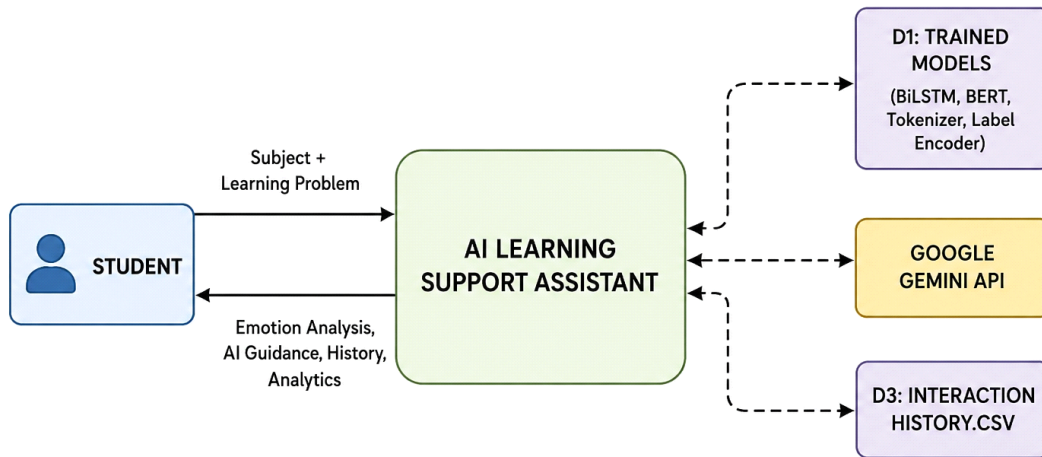
DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Example

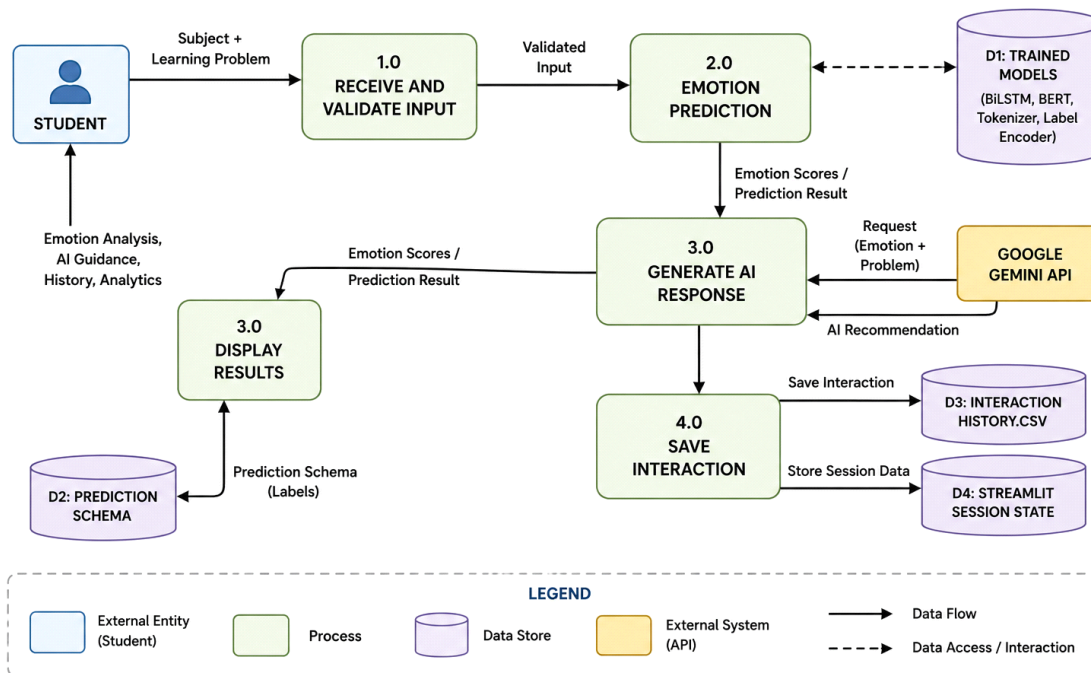
The diagram below illustrates how the symbols above come together to form a complete data flow diagram.

1. CONTEXT DIAGRAM (LEVEL-0 CONTEXT)

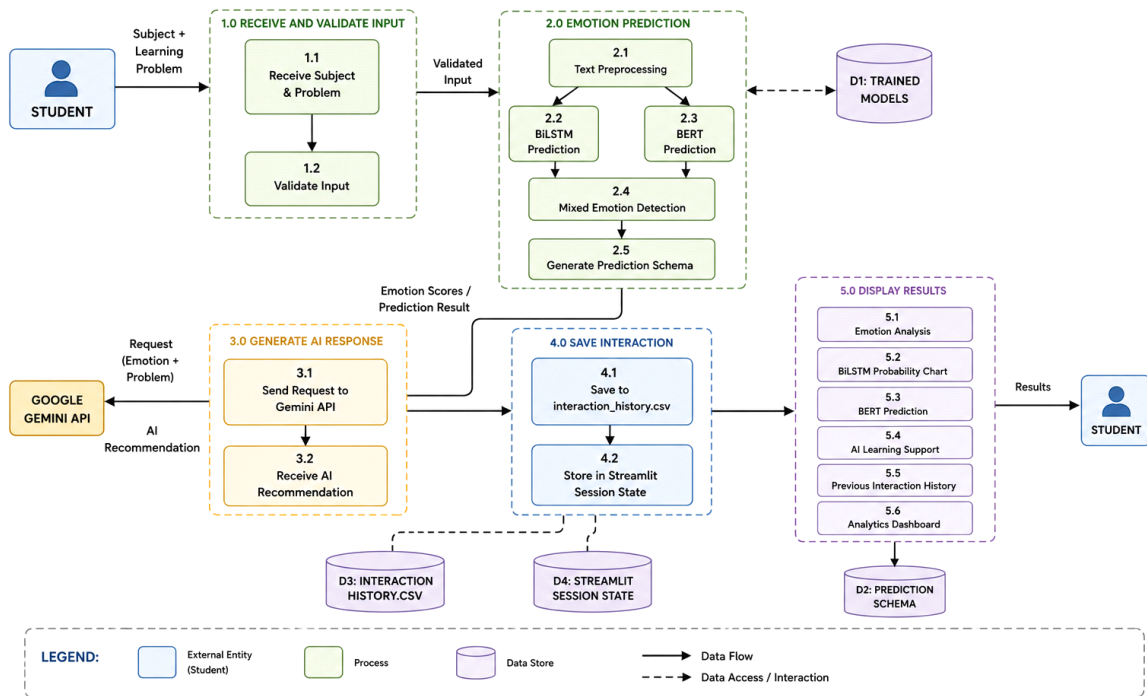


2. LEVEL-1 DATA FLOW DIAGRAM (DFD)

AI LEARNING SUPPORT ASSISTANT



3. LEVEL-2 DATA FLOW DIAGRAM (DFD) AI LEARNING SUPPORT ASSISTANT



4. DFD COMPONENTS AI LEARNING SUPPORT ASSISTANT

Component Type	Component Name / ID	Description	Input	Output	Stored In / Connected To
EXTERNAL ENTITY  Student	Student	The user of the system who submits learning problems and views results, history, and analytics.	- Subject - Learning Problem	- Emotion Analysis - AI Learning Guidance - Previous History - Analytics Dashboard	Interacts with Process 1.0 (Receive User Input)
	1.0 Receive User Input	Receives subject and learning problem from the student and validates the input.	- Subject - Learning Problem	Validated Input	Sends validated input to Process 2.0 (Emotion Prediction)
PROCESSES 	2.0 Emotion Prediction	Predicts emotions using BiLSTM and BERT models and detects mixed emotions.	Validated Input	- Emotion Scores - Prediction Result - Mixed Emotion	Uses D1: Trained Models and updates D2: Prediction Schema
	3.0 Generate AI Response	Generates personalized learning guidance using Google Gemini API.	- Learning Problem - Detected Emotion (Primary + Secondary)	AI Recommendation (Learning Guidance)	Connects with Google Gemini API
	4.0 Save Interaction	Saves the interaction details including predictions and AI response.	- Subject - Problem - Predictions - AI Response - Confidence	Saved Interaction Record	- Stores in D3: interaction_history.csv - Stores in D4: Streamlit Session State
	5.0 Display Results	Displays all results, history, and analytics to the student.	- Prediction Result - AI Response - History Data - Analytics Data	- Emotion Analysis - BiLSTM Chart - BERT Prediction - AI Learning Support - Previous History - Analytics Dashboard	Reads from D2, D3, D4 and displays to Student
DATA STORES 	D1: Trained Models	Stores trained BiLSTM model, BERT model, tokenizer, and label encoder.	–	Provides models for emotion prediction	Connected to Process 2.0 (Emotion Prediction)
	D2: Prediction Schema	Stores standardized prediction results before visualization.	Prediction Result	Standardized Emotion Result	Used by Process 5.0 (Display Results)
	D3: interaction_history.csv	Stores all user interactions permanently in CSV format.	Interaction Record	Saved Interaction History	Used by Process 4.0 and read by Process 5.0
	D4: Streamlit Session State	Stores current session data, recent interactions, and dashboard information.	Session Data	Session Related Data	Used by Process 4.0 and read by Process 5.0
EXTERNAL SYSTEM 	Google Gemini API	Generates personalized learning guidance based on detected emotion and learning problem.	Request (Learning Problem + Detected Emotion)	AI Learning Recommendation	Connected to Process 3.0 (Generate AI Response)

5. DATA FLOW SUMMARY

AI LEARNING SUPPORT ASSISTANT

S.No	Source	Data / Information	Destination	Purpose / Description
1	Student	Subject, Learning Problem	→ 1.0 Receive User Input	User provides subject and learning problem as input to the system.
2	1.0 Receive User Input	Validated Input	→ 2.0 Emotion Prediction	Input is validated and forwarded for emotion prediction.
3	2.0 Emotion Prediction	Emotion Scores / Prediction Result	→ D1: Trained Models	Models (BiLSTM, BERT, Tokenizer, Label Encoder) are used to generate predictions.
4	2.0 Emotion Prediction	Prediction Result	→ D2: Prediction Schema	Prediction results are standardized before visualization.
5	2.0 Emotion Prediction	Emotion Scores / Prediction Result	→ 3.0 Generate AI Response	Prediction results and learning problem are sent to AI to generate personalized guidance.
6	3.0 Generate AI Response	AI Recommendation	→ Google Gemini API	Request is sent to Gemini API and AI recommendation is received.
7	3.0 Generate AI Response	AI Recommendation	→ 4.0 Save Interaction	AI recommendation and prediction details are saved for future reference.
8	4.0 Save Interaction	Interaction Record (Subject, Problem, Emotions, Predictions, AI Response, Timestamp)	→ D3: interaction_history.csv	All interactions are stored permanently in CSV format.
9	4.0 Save Interaction	Session Data (Current Interaction Details)	→ D4: Streamlit Session State	Session related data is stored temporarily for current user session.
10	5.0 Display Results	Emotion Analysis, BiLSTM Chart, BERT Prediction, AI Guidance, History, Analytics	→ Student	All results and analytics are displayed back to the student.

LEGEND: External Entity (Student) Process Data Store External System (API) Data Flow Data Access / Interaction

6. DFD PROCESSES DESCRIPTION

AI LEARNING SUPPORT ASSISTANT

S.No	Process ID & Name	Description	Inputs	Outputs	Key Activities / Functions
1	1.0 RECEIVE USER INPUT	Receives the subject and learning problem from the student and validates the input.	- Subject - Learning Problem	Validated Input	- Receive subject and problem - Validate empty or invalid input - Prepare input for emotion prediction
2	2.0 EMOTION PREDICTION	Predicts emotions using BiLSTM and BERT models and detects primary and secondary emotions.	Validated Input	- Emotion Scores - Prediction Result - Mixed Emotion	- Text preprocessing - BiLSTM emotion prediction - BERT emotion prediction - Mixed emotion detection - Generate prediction schema
3	3.0 GENERATE AI RESPONSE	Generates personalized learning guidance using Google Gemini API based on detected emotion and learning problem.	- Learning Problem - Detected Emotion (Primary + Secondary)	AI Recommendation (Learning Guidance)	- Send request to Gemini API - Receive AI-generated guidance - Return personalized response
4	4.0 SAVE INTERACTION	Saves interaction details including predictions and AI response for future reference.	- Subject - Problem - Predictions - AI Response - Confidence	Saved Interaction Record	- Store data in interaction_history.csv - Store session data in Streamlit Session State
5	5.0 DISPLAY RESULTS	Displays all results, previous history, and analytics dashboard to the student.	- Prediction Result - AI Response - History Data - Analytics Data	- Emotion Analysis - BiLSTM Chart - BERT Prediction - AI Learning Support - Previous History - Analytics Dashboard	- Render emotion analysis - Plot BiLSTM probability chart - Display BERT prediction - Show AI learning support - Load and display previous interactions - Generate analytics dashboard

LEGEND External Entity (Student) Process Data Store External System (API) Data Flow Data Access / Interaction

Data Flow Summary

Source	Data	Destination
Student	Subject, Learning Problem	Receive User Input
Receive User Input	Validated Input	Emotion Prediction
Emotion Prediction	Emotion Scores	Trained Models / Prediction Schema
Prediction Schema	Emotion Result	Gemini AI
Gemini AI	AI Learning Guidance	Save Interaction
Save Interaction	Interaction Record	CSV History
Save Interaction	Session Data	Streamlit Session State
CSV + Session State	History Data	Display Results
Display Results	Emotion Analysis, Charts, AI Response, History, Analytics	Student